**KATHMANDU UNIVERSITY**

**SCHOOL OF ENGINEERING**

**DEPARTMENT OF GEOMATICS ENGINEERING**

FINAL PROJECT REPORT ON

# SOIL EROSION ESTIMATION OF DHADING USING THE RUSLE MODEL

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# ABSTRACT

Soil erosion is a major environmental problem that affects the praffecting land resources' productivity and sustainability attempts to estimate the soil loss of Dhading through the application of the Revised Universal Soil Loss Equation (RUSLE) model. The calculation of erosion requires certain data from various sources available in different formats and scales. Geographic Information System (GIS) allowed considerable time savings in processing spatial data, and screening the effects of each factor affecting soil erosion. Among various erosion factors, topography, rainfall, soil properties, and soil conservation practices were used for the study. Average soil loss was calculated by multiplying these factors. In addition, it analyzes the effect of Land Use and Land Cover (LULC) and slope (β) exposition on soil erosion.

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# LIST OF ABBREVIATIONS

| | |
|---|---|
| DEM | Digital Elevation Model |
| DHM | Department of Hydrology and Meterology |
| ESRI | Environmental Systems Research Institute |
| FAO | Food and Agricultural Organization |
| GIS | Geographic Information System |
| GPS | Global Positioning System |
| LULC | Landuse/ Landcover |
| RUSLE | Revised Universal Soil Loss Equation |
| SOE | School of Engineering |
| SRTM | Shuttle Radar Topography Mission |
| UNESCO | United Nations Educational, Scientific and Cultural Organization |
| USLE | Universal Soil Loss Equation |

# LIST OF SYMBOLS

| Symbol | Description |
|---|---|
| C | Cover management factor |
| CLA | Clay percentage |
| Fcsand | Low soil erodibility factor |
| Fhisand | Factor that reduces soil erodibility for soil with high sand content |
| Forgc | Factor reducing soil erodibility for soil with high organic content |
| Fsi-cl | Low Soil Erodibility with a high clay-to-slit ratio |
| K | Soil erosivity factor |
| L | Slope length factor |
| LS | Topographic factor |
| M | Slope length exponent |
| mT $ha^{-1}$ | Million tonnes per hectare |
| P | Mean annual rainfall in mm |
| R | Rainfall erosivity factor |
| W | Week |
| S | Slope gradient factor |
| SAN | Sand percentage |
| SIL | Silt percentage |
| SN1 | Sand content subtracted from 1 and divided by 100 |
| t $ha^{-1}$ $yr^{-1}$ | Tonnes per hectare per year |
| B | Slope angle (°) |

# 1. INTRODUCTION

## 1.1 Background

Soil erosion is a significant cause of soil degradation and is closely linked to agricultural intensification, deforestation, and unsustainable human activities (Roslee & Sharir, 2019). According to Wubalem (2022), water-induced soil erosion is a critical natural degradation problem that results from the impact of rainfall and runoff, which removes fertile topsoil and diminishes cultivable land, leading to a decrease in food production. Additionally, eroded sediments degrade water quality and contribute to the eutrophication of freshwater bodies (Pimentel, 2006). Soil erosion is influenced by various factors such as soil, rainfall, terrain, and land cover (Kayet, Pathak, Chakrabarty, & Sahoo, 2018).

In Nepal, approximately 45.5% of the total land is affected by water erosion, particularly through sheet and rill erosion (Koirala, Thakuri, Joshi, & Chauhan, 2019). Studies conducted in Nepal have shown that soil loss through surface erosion from agricultural land in the hills varies from less than 2 tonnes per hectare per year to as high as 105 tonnes per hectare per year (Chalise, Kumar, & Kristiansen, 2019). For example, a study conducted in 1992 and 1993 along the Likhu Khola River revealed that the rate of soil loss was 11 tonnes per hectare per year under grassland and slightly degraded secondary forest, 2.7–8.2 tonnes per hectare per year under rain-fed cultivation, and 2 tonnes per hectare per year under no cultivation (Shrestha, n.d.). The RUSLE model with GIS is commonly used for estimating the spatial distribution of soil erosion but has limitations in predicting gully erosion (Wang et al., 2013). Despite this limitation, the RUSLE model guides the development of conservation plans to control erosion under different land-cover conditions, such as croplands, rangelands, and disturbed forest lands (Millward & Mersey, 1999).

Various models exist to predict water-induced erosion, ranging from empirical (USLE/RUSLE) to physical or process-based models, and they differ in complexity and data input requirements. The Revised Universal Soil Loss Equation (RUSLE) is widely used for soil erosion assessment, as it takes into account factors such as climate, soil, topography, and land use, and provides a

framework for erosion prediction at different spatial scales (Renard et al., 1997). The RUSLE model computes the average annual erosion by multiplying factors such as rainfall erosivity (R), soil erodibility (K), slope length and steepness (LS), cover management (C), and support practice (P) (Koirala et al., 2019).

## 1.2 Problem Statement

The research area's potential concern, soil erosion, could have a significant impact on both the study region (Dhading) and the surrounding environment. The most prominent problems related to soil erosion in this area are sedimentation in lowland areas and the emergence of mudflows at higher elevations. Changes in stream characteristics, such as drainage depth and water flow velocity, can lead to increased sediment deposition, which would impact the hydrological system of the stream. In extreme circumstances, the lower ground in the area could experience flash floods.

According to the survey conducted by Giri & Dahal (2021) regarding climate change and agriculture in Dhading, the soil of the study area was found to be inherently fragile and susceptible to degradation, with wind, landslides, and water runoff causing significant topsoil loss during the pre-monsoon and monsoon seasons. While water-induced soil erosion is considered a major land degradation process, the spatial variability of soil susceptibility to erosion in the study area has not been assessed and mapped. Therefore, the study aims to produce important results on erosion and soil loss data by assessing the extent and mapping soil erosion risk areas using GIS with the RUSLE model. This study can identify areas at potential risk of extensive soil erosion in the study area and provide a framework for decision-makers to plan erosion control activities. Additionally, it can help fill the gap in soil loss information for a particular district and serve as a source of information for future researchers interested in similar research.

## 1.3 Objectives

The primary objective of soil erosion estimation of Dhading using the RUSLE model.

The secondary objectives include:

i. Analyze soil erosion rates based on slope and land use/land cover.
ii. Categorization of the area into different erosion classes based on the severity of erosion.
iii. Generation of individual factors map.

## 1.4 Scope

The project will involve a comprehensive study to estimate soil erosion in Dhading, a district in Nepal, using the RUSLE (Revised Universal Soil Loss Equation) method integrated with Geographic Information System (GIS) tools. The study will start with selecting Dhading as the study area due to its diverse topography and significant agricultural activities. Relevant data will be collected, including rainfall records, soil properties, topographical details, land cover information, and data on support practices within Dhading.

The RUSLE model will be used to quantify soil erosion, considering factors such as rainfall-runoff erosivity, soil erodibility, slope length and steepness, cover management, and support practices. GIS tools will be used to spatially analyze these factors and create detailed maps showing the distribution of soil erosion risk across Dhading. The analysis will focus on identifying areas with high erosion potential and evaluating the impact of various land uses and practices on soil erosion rates within the district.

The results will be interpreted to provide insights into the spatial patterns of soil erosion in Dhading, comparing the estimated soil loss with acceptable limits. Based on the findings, the project will suggest soil conservation measures and management practices to mitigate erosion and provide policy recommendations for sustainable land use planning in Dhading. Finally, the outcomes, analyses, and recommendations will be compiled into a comprehensive final report documenting the entire study and its conclusions.

# 2. LITERATURE REVIEW

Timely and accurate estimation of soil erosion loss or risk assessment has become essential in many countries to estimate how quickly soil is eroded before the conservation strategy is compromised. Based on the complexity of the models, the physical processes they imitate, the equations they utilize, and the accuracy they aim for, different simulation models empirical, stochastic, and deterministic (either physically based or conceptual) were developed in response to this demand (Efthimiou et al., 2014).

The Universal Soil Loss Equation (USLE), created by Wischmeier and Smith (1978) based on the lumped approach to catchment scale, is one of the most extensively used empirical models because of its relative resilience, simplicity, and representation of a standardized methodology. Additionally, it is thought to be the best technique for estimating soil loss in the absence of measurements. Since the initial USLE publication, more research, tests, data, and resources have become accessible, and a deeper understanding of the erosion process has all contributed to the development of the Revised Universal Soil Loss Equation (Renard et al., 1991). The equation's essential structure has not changed; however, changes have been made to some of the factors. New and revised changes (the R factor broadened to encompass the Western US) are among these modifications (filling of data gaps and a more refined smoothing of the R iso-erodent map of the Eastern U.S. has also been performed). The slope length (L) and steepness (S) factors have been revised to account for the relationship between rill and inter-rill erosion, iso-erodent maps and corrections made to account for rainfall on ponded water, a time-varying (seasonal) approach for the soil erodibility factor (K), a new equation to reflect slope length and steepness, a sub-factor approach for evaluating the cover-management factor (the C factor no longer represents seasonal soil-loss ratios The factor P has been modified to incorporate conditions for rangelands, contouring, strip-cropping, and terracing. It now reflects a continuous function of previous land use, surface cover, crop canopy, surface roughness, and soil moisture (Renard et al., 1991).

The typical approach to implementing the RUSLE is to estimate each of the model's factors, which is based on several literatures. Previous researchers have devised several methods for

predicting these variables, including the use of climate data, soil and geological maps, remotely sensed satellite images, empirical formulas, and digital elevation models (DEM). The average soil 5 loss (A) due to water erosion per unit area per year (Mg $ha^{-1}$ per year) was quantified, using RUSLE by the following equation:

$$A = R \times K \times L \times S \times C \times P$$

Where, A=soil loss (t $ha^{-1}$ $yr^{-1}$), R=rainfall erosivity factor (mm $ha^{-1}$ $yr^{-1}$), K=soil erodibility factor (mg $ha^{-1}$ $yr^{-1}$), LS=slope-length and slope steepness factor (dimensionless), C=land management factor (dimensionless), and P=conservation practice factor (dimensionless).

# 3. METHODOLOGY

## 3.1 Study Area

The study area is Dhading district, Nepal, located between the coordinate of latitude: 27.9667° N and the coordinate of longitude: 84.8667° E in the mid-hilly part of Nepal. It is extended in an area of 1905 km². It is located in the central development region of Nepal with an elevation range from 300 meters to 7409 meters. It is bordered to the east by Nuwakot and Kathmandu, west by Gorkha, north by Rasuwa, and south by Makawanpur and Chitwan.

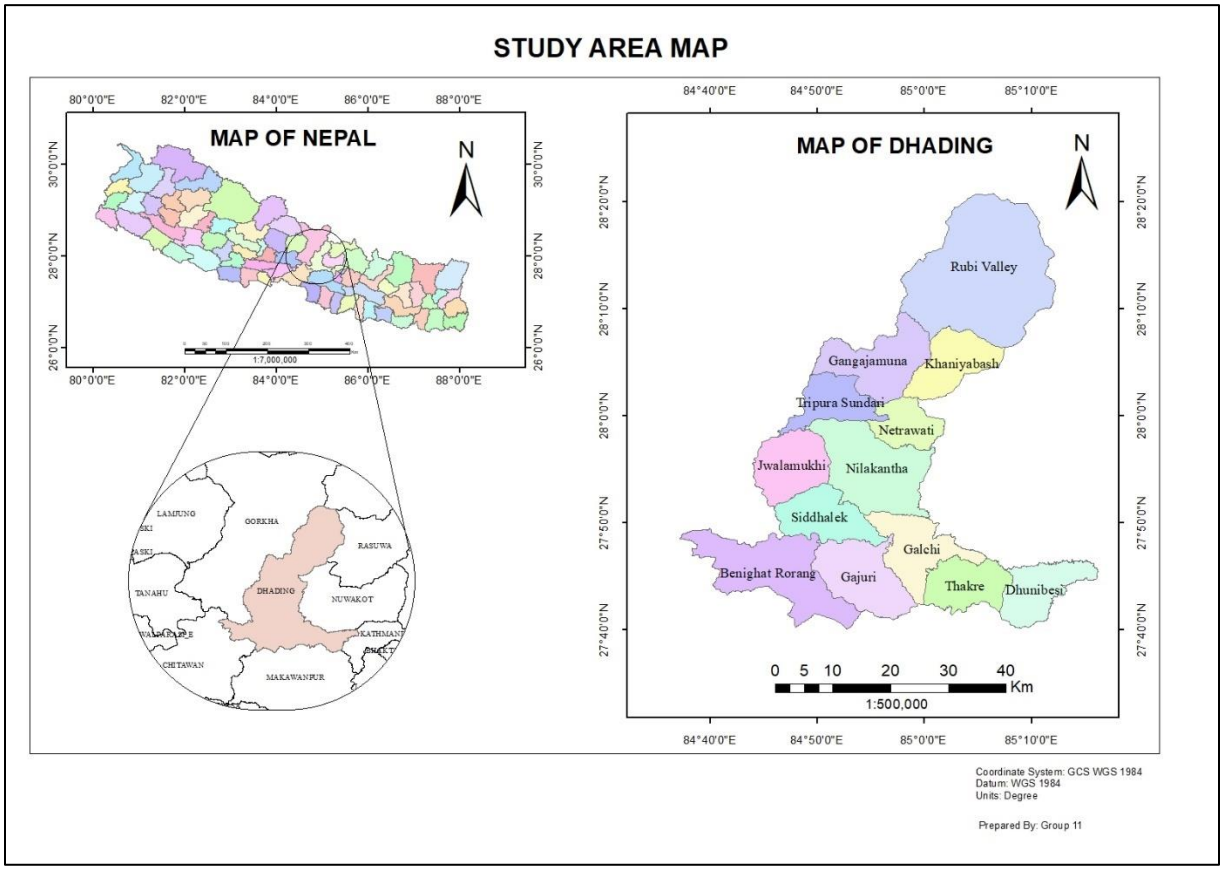


**Figure 1: Study Area Map**

## 3.2 Study Method/Design/Workflow

The following input data are required as GIS datasets: average annual precipitation (dataset), digital soil map with information regarding the topsoil layer, Digital Elevation Model (DEM), digital land use data about land use classes, and objects that inhibit erosion (barriers). Once provided with this set of data, the RUSLE model links erosion factors influencing soil erodibility (K factor), erosivity (R factor), land cover and management (C factor), slope length (L factor), and slope gradient (S factor). By multiplying these factors, the mean relative soil loss in tons per hectare per year can be calculated. The flowchart showing the detailed procedures and data needed are summarized in Figure 2.

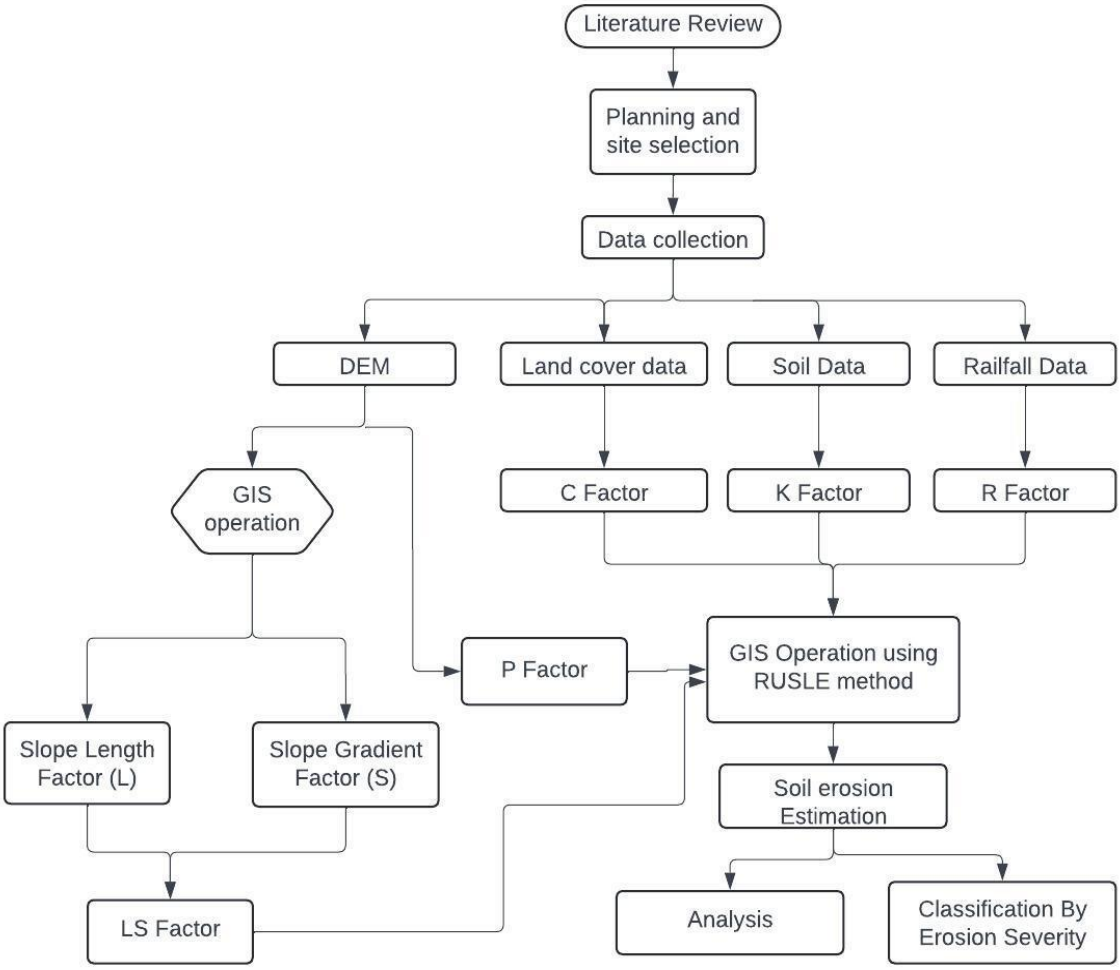


**Figure 2: Workflow**

## 3.3 Data Sources to Be Used

This study used various datasets developed from different sources. Applied datasets and their relevant sources are listed below

**Table 1: Data Sources**

| Data sets | Data Sources | Resolution |
|---|---|---|
| DEM | https://asf.alaska.edu/datasets/daac/alos-palsar/ | 12.5 m |
| Soil Data | FAO/UNESCO Soil Map of the World | 250 m |
| LULC | ESRI Landsat and Sentinel | 10 m |
| Rainfall Data | https://www.dhm.gov.np/ | Vector data |

## 3.4 Software/Instruments to Be Used

The laptop was used as hardware for this project. ArcGIS 10.8 software was used for the mapping and analysis procedure.

## 3.5 Description of Soil Erosion Parameters

### 3.5.1 Rainfall Erosivity Factor (R)

The rainfall erosivity factor (R) describes the erosivity of rainfall at a particular location based on the rainfall amount and intensity. It reflects the effect of rainfall intensity on soil erosion. The rainfall erosivity used in the RUSLE must quantify the effect of raindrop impact and explain the amount and rate of runoff associated with rainfall (W.H Wischmeie, 1978) and its unit is expressed in MJ mm $ha^{-1}$ $yr^{-1}$. The rainfall map represents the mean annual precipitation over the country, produced from the ground meteorological stations. The equation integrated to generate the R-factor is given by reference (Morgan, 2009).

$R = 38.5 + 0.35P$,

where, $R$ = Rainfall Erosivity Factor, $P$ = Mean Annual Rainfall in mm

### 3.5.2 Soil Erodibility Factor (K)

The soil erodibility factor (K) is a quantitative description of the inherent erodibility of a particular soil type; it is a measure of the susceptibility of soil particles to detachment and transport by rainfall and runoff. The main soil properties influencing the K factor are soil texture, organic matter, soil structure, and permeability of the soil profile. For a particular soil, the soil erodibility factor is the rate of erosion per unit erosion index from a standard unit plot of 22.13 m long slope length with a 9% slope gradient (B.P. Ganasri, 2016). It reflects the rate of soil loss per rainfall erosivity (R) index.

Organic carbon= (1-((0.0256* [OC_topso])/( [OC_topso] +Exp ( 3.72-(2.95 * [OC_topso])))))

Fraction of silt= (1-((-0.7*(1- [sand_top]/100))/((1 - [sand_top]/100) +Exp ( -5.51 +22.9 * (1- sand_top]/100)))))

Fraction of Clay= ( [silt_top] /( [clay_top]+ ([silt_top])))^0.3

Fraction of Sand = (0.2+(0.3*Exp ((-0.256 * [sand_top] *(1- [silt_top]/100 )))))

K_RUSLE= [Organic carbon]*[Fraction of Sand]*[Fraction of silt]*[Fraction of Clay] *0.1317

### 3.5.3 Topographic Factor (LS)

The topographic factor or Slope Length and Steepness Factors (LS) were created from two sub-factors: a slope gradient factor (S) and a slopelength factor (L); both of which are determined from the Digital Elevation Model (DEM). Slope length and gradient is the important parameter in soil erosion modeling (Koirala et al., 2019), in calculating the transport capacity of overland flow (Surface runoff). The L represents the effect of slope length on erosion and the S represents the effect of slope steepness on erosion (Ganasri and Ramesh, 2016). Therefore, the soil loss per unit area increases as the slope length increases but this loss increases more rapidly with slope steepness than it does with slope length. The slope gradient and slope length factors were calculated from the DEM and combined to the result in the topographical factor grid, using the following equations (Gao et al. 2012): $L= (\frac{\lambda}{22.13})^m$

where, L = slope length factor, λ = slope length (m), m = slope-length exponent

$m = \frac{F}{(1+F)}$ ,

F= sinβ 0.0896 /3(sinβ) 0.8 + 0.56

where, F = Ratio of rill erosion to interrill erosion, β = slope angle (◦)

In ArcGIS, L is calculated as:

$L = (flowacc+625)^{(m+1)} – flowacc^{(m+1)}/ 25^{(m+2)}) * 22.13m$

For the slope gradient factor, the calculation is as:

S =Con((Tan(slope*0.01745) < 0.09), (10.8 * Sin(slope * 0.01745) + 0.03), (16.8 * Sin(slope *0.01745)− 0.5))

Final computation with LS = L∗ S,

### 3.5.4 Cover-management factor (C)

The cover-management factor (C) is used to reflect the effect of cropping and other management practices on erosion rates. Vegetation cover is the second most important factor next to topography that controls soil erosion risk. The land cover intercepts rainfall, increases infiltration, and reduces rainfall energy. In areas where land uses other than cropping dominate, as in the Himalaya, the C factor is normally assigned based on a simple assessment of vegetation cover, rather than close analysis of agricultural cropping patterns. In this study, Land Use/Land Cover (LULC) produced by the ESRI Landsat and Sentinel was used for preparing a C-factor map. First, the raster map was converted to a polygon and the attributes with the same landuse type were merged in ArcGIS. From this, eight types of landuse were obtained (Table 2). For each landuse type, C values were assigned through reference (P Panagos, 2015). The C factor ranges from 0 to approximately 1, where higher values indicate no cover effect and soil loss comparable to that from a tilled bare fallow, while lower C means a very strong cover effect resulting in no erosion (Z Erencin, 2000)

**Table 2: C Factor**

| **Land Use** | **C factor** |
|---|---|
| Forest | 0.03 |
| Shrubland | 0.03 |
| Grassland | 0.01 |
| Agricultural Land | 0.21 |
| Barren Land | 0.45 |
| Water Body | 0.00 |
| Snow Glacier | 0.00 |
| Built-Up | 0.00 |

### 3.5.5 Practice factor (p) `

Practice factor indicates the rate of soil loss across various cultivated lands. This factor can control erosion through methods such as contours, cropping, and terracing (G.J.,1999). The P values for support practices range

from 0 to 1, where 0 represents very good resistance to anthropogenic erosion and 1 indicates a lack of resistance to erosion. In Nepal, farming practices on sloped agricultural land often involve constructing terraces similar to contour farmland, which is a form of conservation farming. Consequently, we consider contour farmland as a form of agricultural support practice.

**Table 3: P Factor**

| Slope (%) | Contouring |
|---|---|
| 0-7 | 0.55 |
| 7-11.3 | 0.60 |
| 11.3-17.6 | 0.8 |
| 17.6-26.8 | 0.95 |
| <26.8 | 1 |

# 4. PROJECT MANAGEMENT

## 4.1 Logical Framework Analysis

**Table 4: Logical framework Analysis**

| S.N. | Activities | Risk | Who | When | How | Remedies |
|---|---|---|---|---|---|---|
| 1. | Literature Review | Difficult Organize | All | Until Data Analysis | Article study | Use Google Scholar and another site |
| 2. | Site Selection | Functionality of the model in our study area | Prem Khadka, Sagar Upadhaya | Week 1 | Discussion with supervisors | Supervisor's recommendation |
| 3. | Concept Paper submission | | Bibechana Khadka | Week 3 | Hardcopy and softcopy submission | |
| 4. | Proposal Defense | | All | Week 4 | Presentation | |
| 5. | Data Collection | All data may not be obtained | All | Week 6,7 | Collection required Dataset | Using different Data sources |
| 6. | Data Analysis | Dealing with some new tools | All | Week 8,9,10 | Using arc GIS | Doing study research on GIS tools |
| 7. | Outcome | | All | Week 12 | Final report presentation and question/ANS session | |

# 5. BUDGET

In our project data are collected from various sources. These are available on the website for free, as opposed to rainfall data. Other stationery and printing costs are only our project expenses. Which are estimated below.

**Table 5: Budget**

| S.N. | Description | Cost |
|---|---|---|
| 1. | Stationery cost | 500 |
| 2. | Printing cost | 800 |
| 3. | Rainfall Data | 400 |

# 6. RESULT

## 6.1 Factor maps

The results showed that the Rainfall Erosivity Factor (R) value ranges between 543.593 and 1043.97 MJ mm $ha^{-1}$ $yr^{-1}$ with the highest values being in Khaniyabas and Benighat rorang Rural municipality and the lower values in the Dhunibesi municipality. The soil Erodibility Factor (K) value was 0.0198 in the northern part of the district and 0.020282 in the southern. The topographic Factor (LS) value for the entire area ranged from 0 to 50.1656. The Cover Management Factor (C) value ranged between 0 and 0.45. The value indicates the percentage erosive capacity in comparison with the bare fallow area. The highest value of 0.45 indicates that 45% of erosion occurs in the area compared to the bare fallow land. The Support Practice Factor (P) value ranged from 0.55 to 1 where a higher value indicates there is no support practice such that erosion is at its maximum due to the absence of any practice.

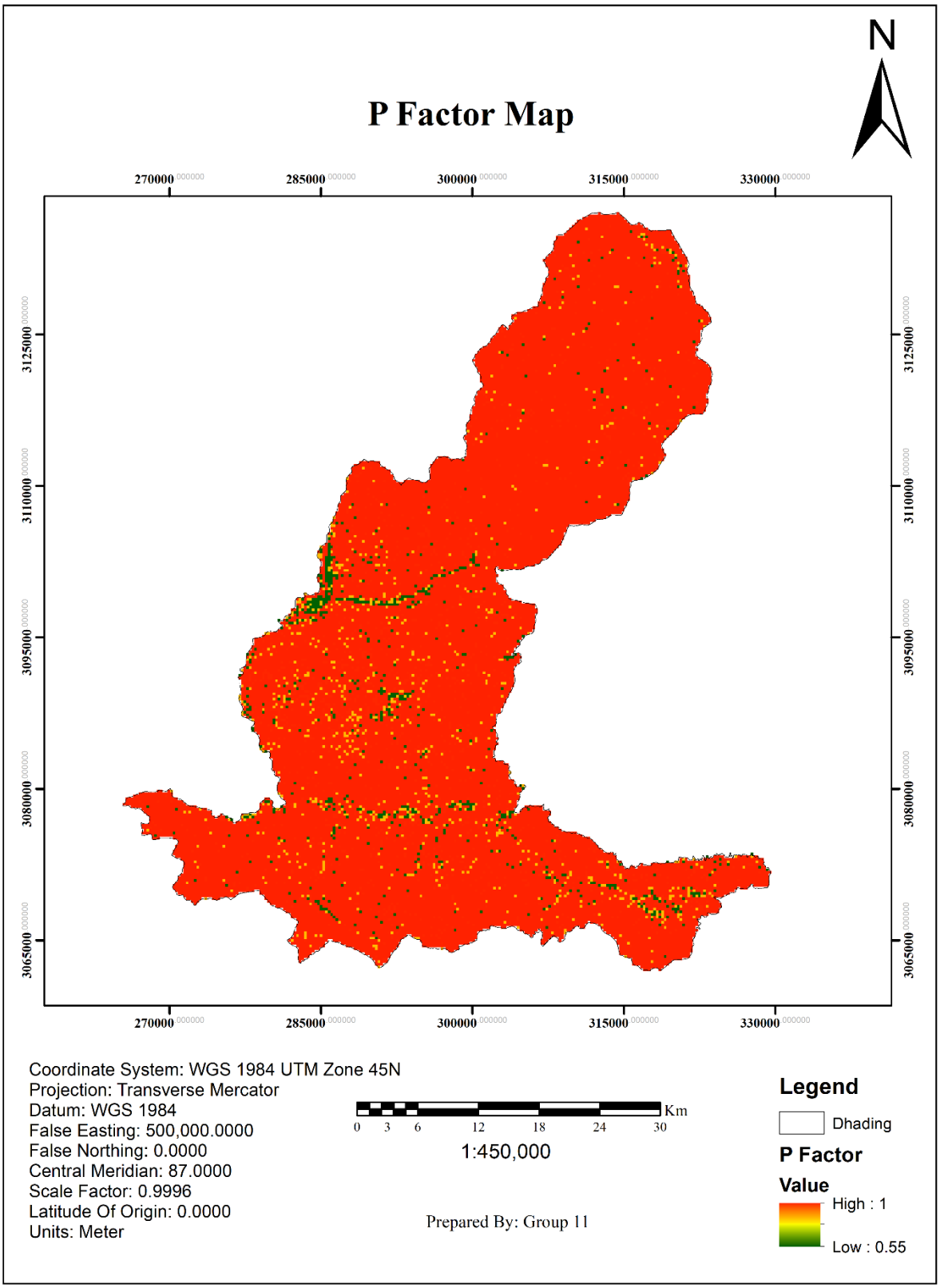


**Figure 3: P Factor Map**

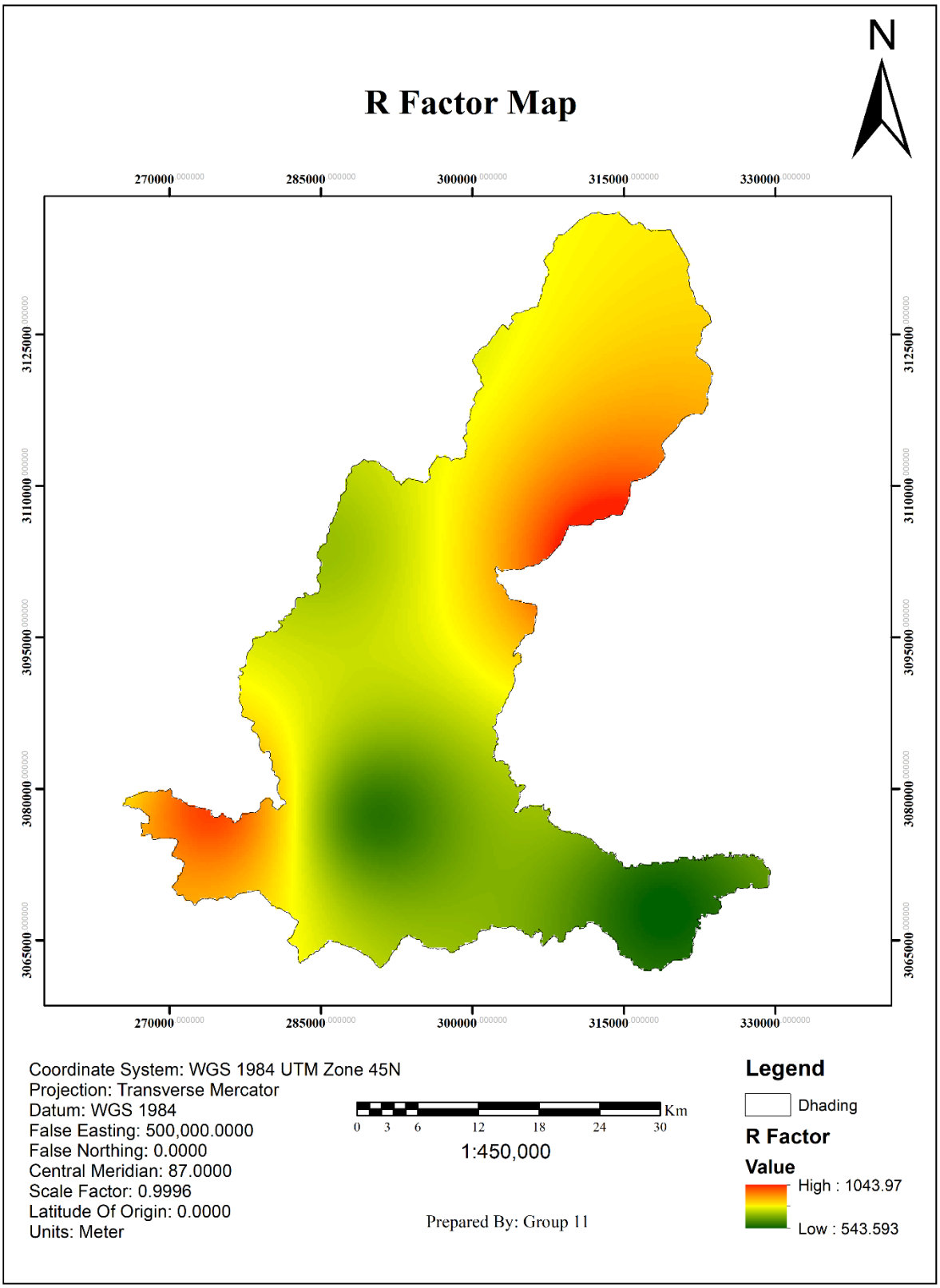


**Figure 4: R Factor Map**

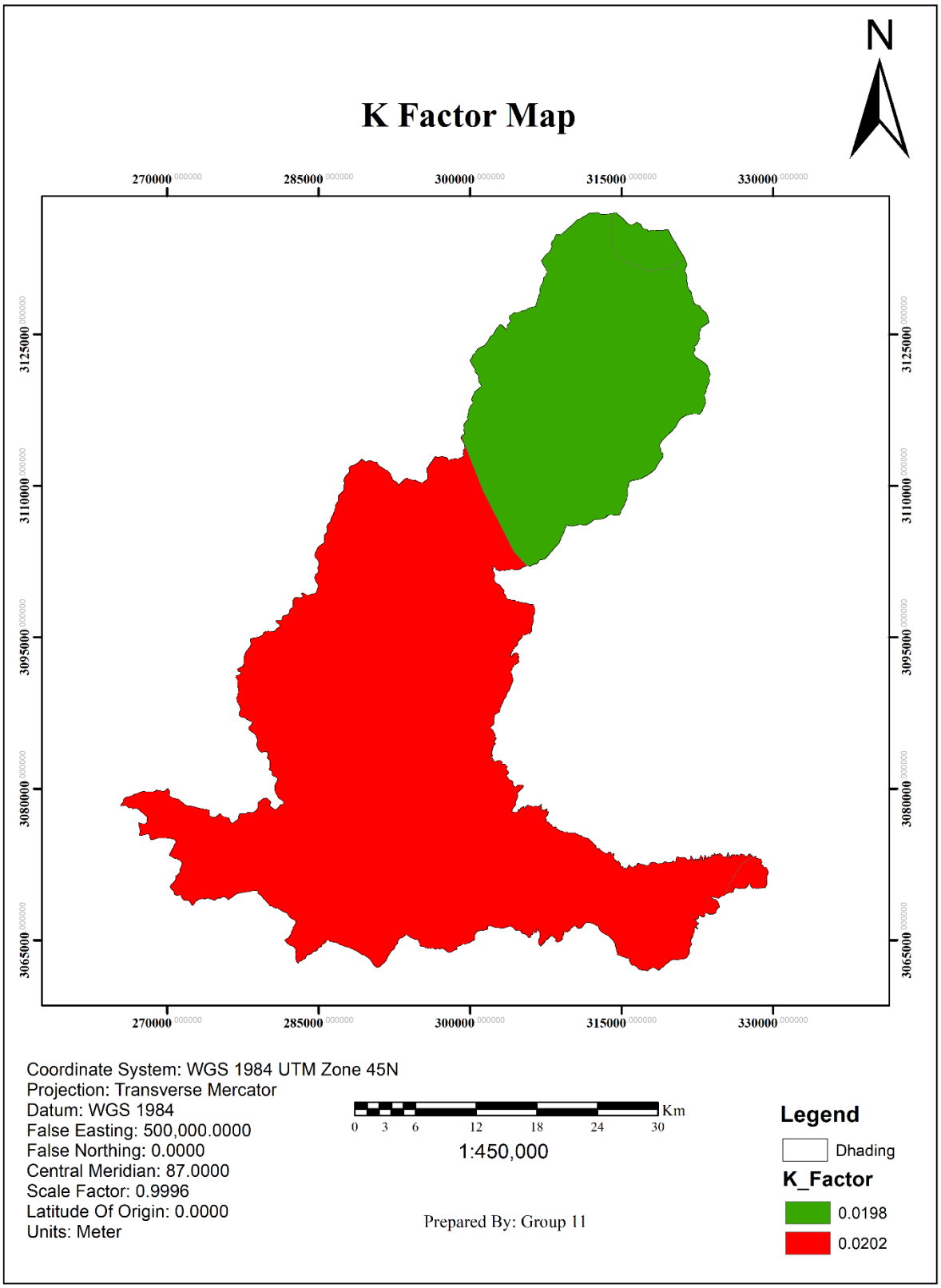


**Figure 5: K Factor Map**

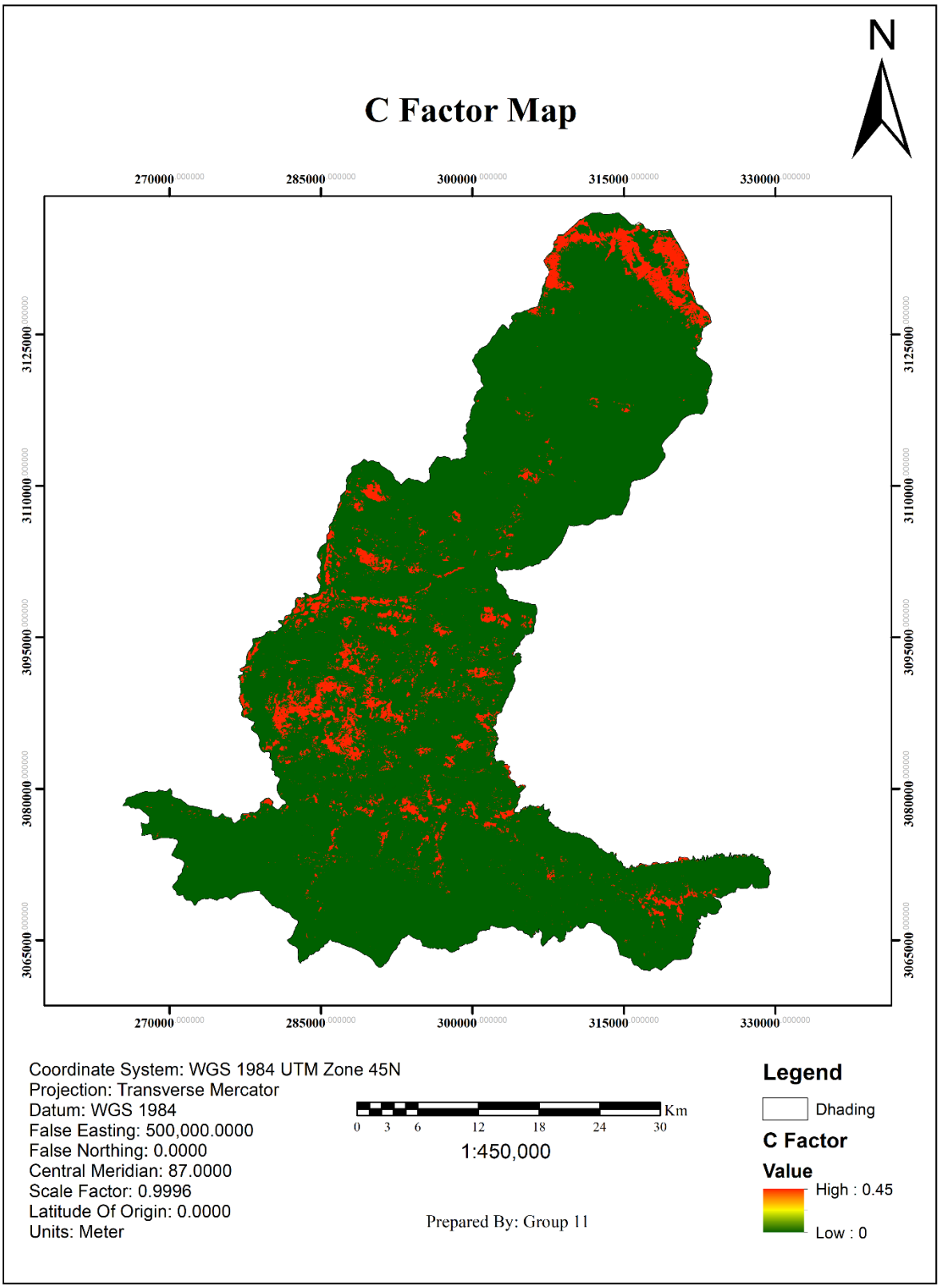


**Figure 6: C Factor Map**

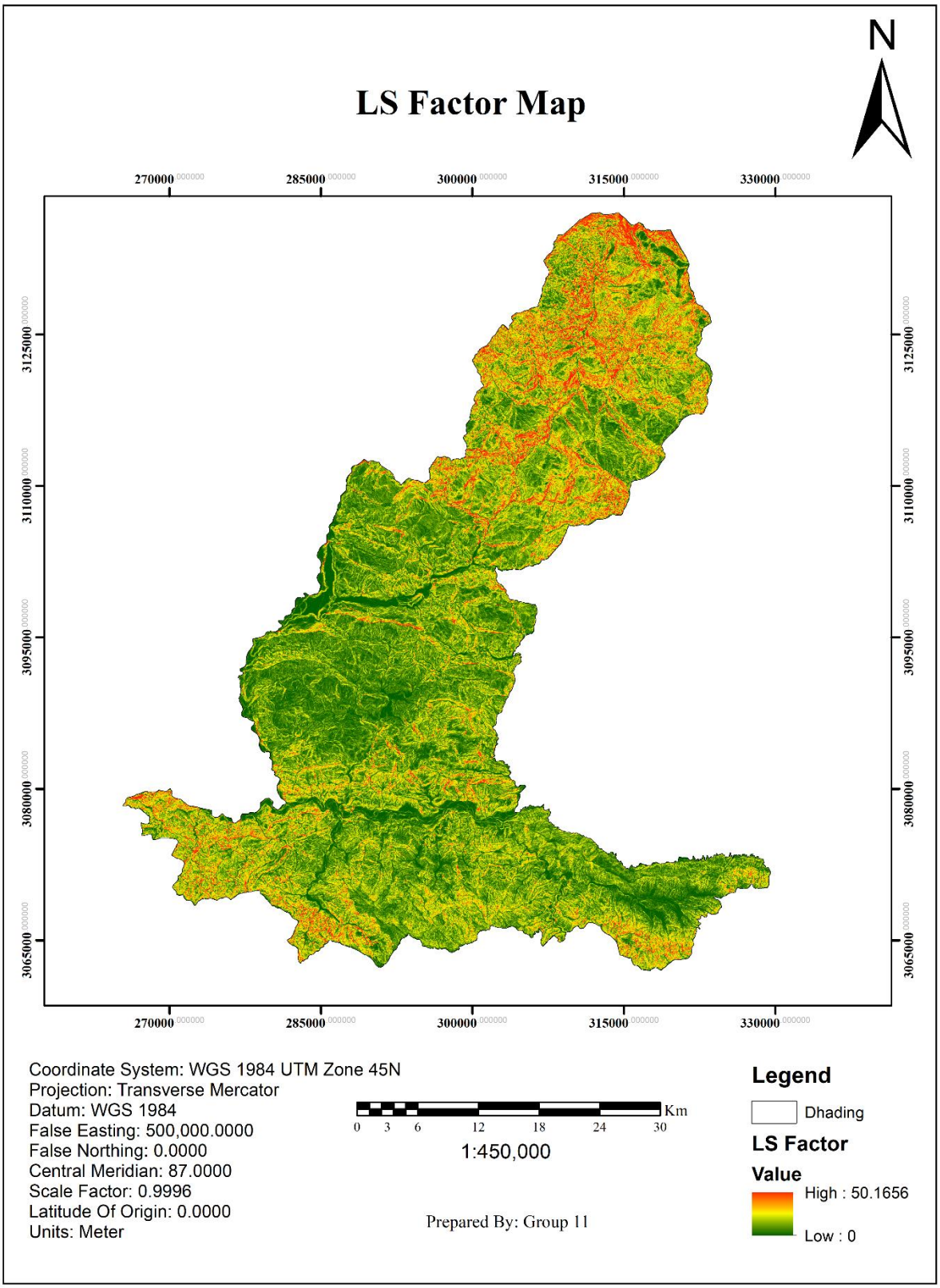


**Figure 7: LS Factor Map**

## 6.2 Potential Soil Erosion Rate of Dhading

The potential soil erosion map of Dhading has been produced in ArcGIS by multiplying the factor maps. It has been found that the erosion ranges from 0 to 364.295 t $ha^{-1}$ $yr^{-1}$ for the entire study area and classification by erosion rate is shown by table 6 and map in figure. The total annual mean soil loss has been estimated to be 10.4 million tonnes for all of Dhading.

**Table 6: Potential Soil Erosion Rate**

| **Erosion Class ($t$ $ha^{-1}$ $yr^{-1}$)** | **Area (hectare)** | **Percentage (%)** | **Mean Erosion ($t$ $ha^{-1}$ $yr^{-1}$)** | **Total Erosion ( *million tonnes* $yr^{-1}$ )** |
|---|---|---|---|---|
| 0-5 | 138259.856 | 73.31 | 2.054467 | 2.8405 |
| 5-10 | 35290.215 | 18.71 | 6.89779 | 2.4342 |
| 10-20 | 10546.725 | 5.59 | 12.924813 | 1.3631 |
| 20-40 | 1395.490 | 0.73 | 27.216229 | 0.3798 |
| 40-80 | 1173.351 | 0.62 | 58.593828 | 0.6875 |
| >80 | 1918.393 | 1.01 | 140.522788 | 2.6957 |
| **Total** | | **100** | | **10.40** |

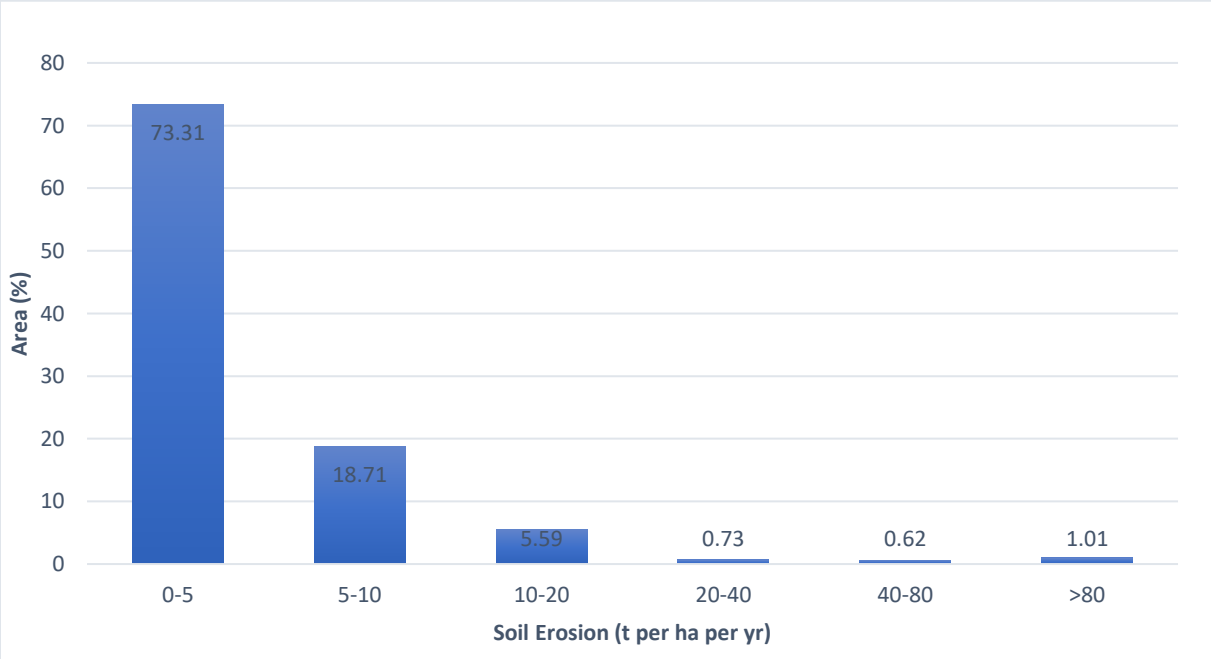


**Figure 8: Soil erosion percentage**

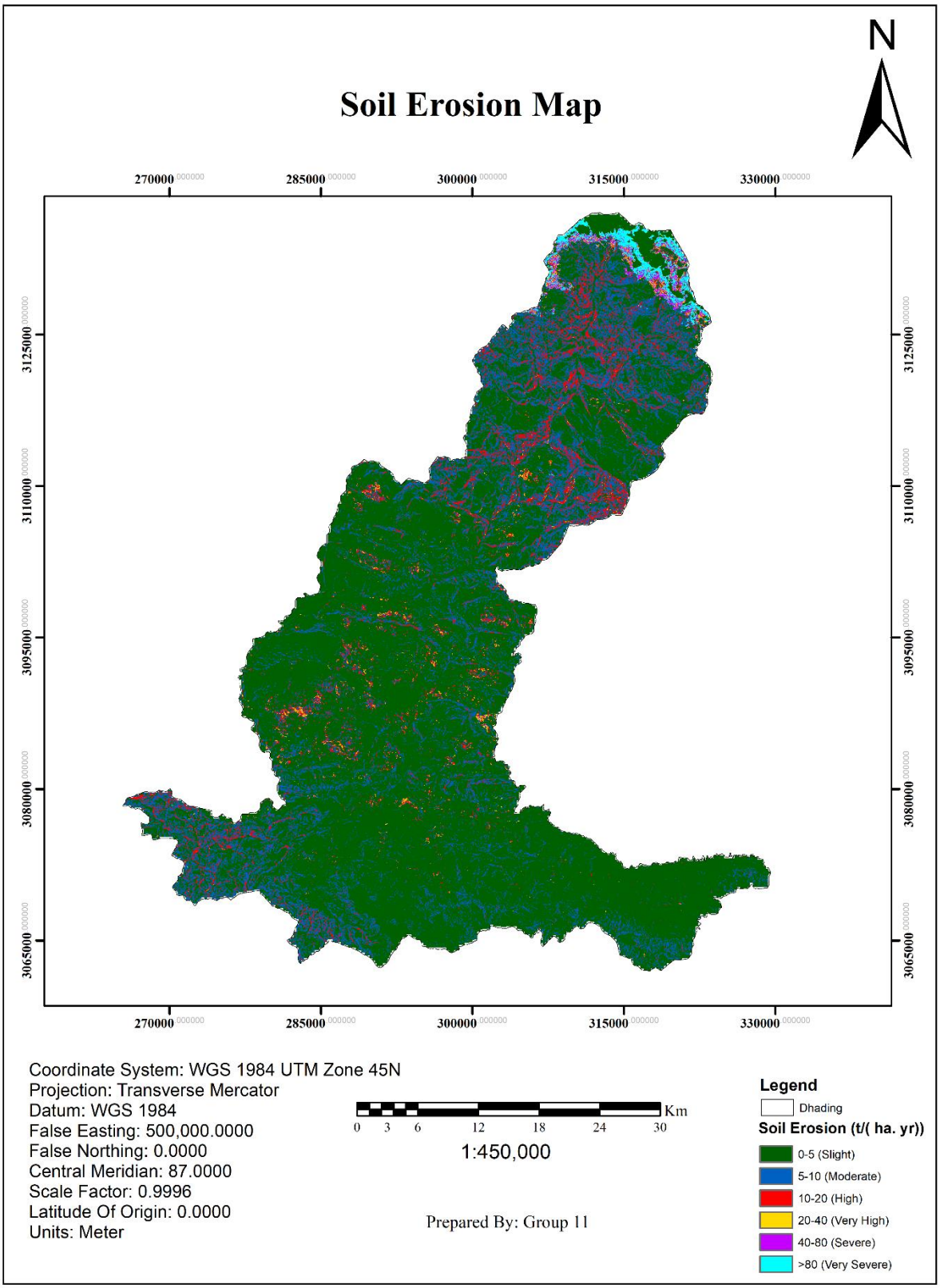


**Figure 9: Soil erosion map**

## 6.3 Classification by Severity and Conservation Priority

The soil erosion rate of Dhading has been classified into 6 erosion classes, indicating the conservation priority regions based on the erosion severity as classified by Singh (1992). The locations with extremely high erosion values have been identified as higher priority. According to the report, only 1.01 % of the regions require instant conservation measures due to their extremely high rates of erosion.

**Table 7: Classification By Severiy and Conservation Priority**

| **Erosion Rate ($t$ ha$^{-1}$ yr$^{-1}$)** | **Class** | **Area (hectare)** | **Percentage (%)** | **Conservation Priority** |
|---|---|---|---|---|
| 0-5 | Slight | 138259.856 | 73.31 | 1 |
| 5-10 | Moderate | 35290.215 | 18.71 | 2 |
| 10-20 | High | 10546.725 | 5.59 | 3 |
| 20-40 | Very high | 1395.490 | 0.73 | 4 |
| 40-80 | Severe | 1173.351 | 0.62 | 5 |
| >80 | Very Severe | 1918.393 | 1.01 | 6 |

# 7. DISCUSSION AND CONCLUSION

In this study, the Revised Universal Soil Loss Equation (RUSLE) model was used to estimate soil erosion in Dhading. Geographic Information System (GIS) tools were employed to process spatial data and evaluate the impact of various factors such as topography, rainfall, soil properties, and land cover. The analysis revealed significant spatial variability in soil erosion rates, with higher potential for erosion in areas characterized by high rainfall erosivity, steep slopes, and low vegetation cover.

Our study found significant variation in soil erosion rates across Dhading, with estimated annual soil loss ranging from 0 to 364.295 tons per hectare per year (t $ha^{-1}$ $yr^{-1}$). The highest erosion rates are observed in the top northern and southern parts of the district, particularly in Rubivalley Rural Municipality, Khaniyabas Rural Municipality and Benighat Rorang Rural Municipality, where rainfall erosivity (R) values range between 543.593 and 1043.97 MJ mm $ha^{-1}$ $yr^{-1}$, as shown in the R factor map (Figure 4). This high rainfall intensity, combined with steep slopes (LS values up to 50.1656, Figure 7), significantly contributes to soil erosion in these regions.

The P factor map (Figure 3) illustrates those areas with lower P values, indicating effective soil conservation practices such as terracing, which are concentrated in agricultural areas with slopes less than 7%. Conversely, higher P values are found in regions with slopes greater than 26.8%, indicating minimal soil conservation efforts. This lack of support practices exacerbates erosion in these areas.

Soil erodibility, represented by the K factor map (Figure 5), shows lower K values in the northern part of Dhading, suggesting that soils in these areas are less prone to erosion. The southern part, however, exhibits higher K values, indicating more erodible soils. This, combined with high rainfall erosivity and steep slopes, results in severe soil erosion.

The C factor map (Figure 6) highlights the impact of land cover on soil erosion. Forested areas have low C values (0.03), indicating high vegetation cover and low erosion risk. Agricultural

lands have higher C values (up to 0.21), reflecting less protective cover and higher susceptibility to erosion. Barren lands have the highest C values (0.45), indicating minimal vegetation and high erosion potential.

Comparing our findings with previous studies, such as (Koirala, Thakuri, Joshi, & Chauhan, 2019), which reported similar high erosion rates in Nepal's hilly regions, validate our results. Our estimated annual soil loss of 10.4 million tons for Dhading aligns with national estimates, where soil loss ranges widely from 2 to 105 t $ha^{-1}$ $yr^{-1}$ depending on land use and topography. This consistency highlights the reliability of using the RUSLE model with GIS tools for soil erosion assessment.

The soil erosion map (Figure 8) synthesizes the combined effect of all RUSLE factors, showing the spatial distribution of potential soil erosion rates across Dhading. Areas with the highest erosion rates (over 80 t $ha^{-1}$ $yr^{-1}$) are primarily located in regions with high R, K, and LS values, highlighting the critical need for soil conservation measures in these high-risk areas.

The study faced limitations, including the resolution of the dataset used for land cover classification, which may have introduced some inaccuracies. For instance, the soil erodibility factor (K) ranges from 0.0198 to 0.020282, which may not fully capture the spatial variability of soil properties across Dhading. Additionally, the RUSLE model's assumption of uniform rainfall distribution may not accurately represent local variations, potentially affecting the precision of erosion estimates, especially in areas with significant microclimatic differences.

Our findings underscore the urgent need for soil conservation measures in Dhading, particularly in high-risk areas. For example, areas with soil erosion rates exceeding 80 t $ha^{-1}$ $yr^{-1}$, comprising 1.01% of the study area, should be prioritized for conservation efforts. Implementing targeted soil conservation practices, such as reforestation, contour farming, and terracing, can significantly mitigate erosion. Policymakers can use these results to effectively plan and implement soil conservation measures, ensuring sustainable land use and agricultural productivity.

Future research should focus on integrating higher-resolution data and advanced remote sensing techniques to improve the accuracy of erosion estimates. Field-based validation of RUSLE estimates through direct soil loss measurements can enhance model accuracy. Additionally, exploring the combined use of RUSLE with other models, such as the WEPP (Water Erosion Prediction Project), could provide more comprehensive insights. Expanding the study to include seasonal variations in rainfall and erosion and long-term monitoring would also be beneficial.

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# 9. ANNEXES

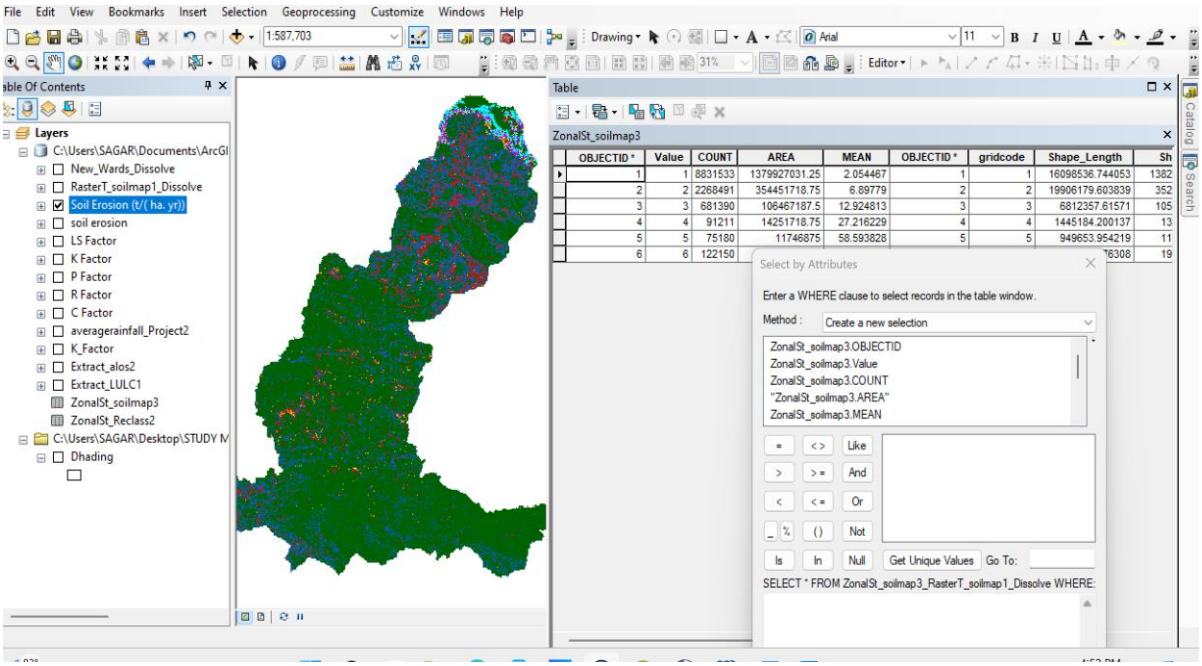


**Figure 10: Calculation of soil erosion**

**Table 8: Annual rainfall data**

| Station Index | Name | District | Latitude (degree) | Longitude (degree) | Elevation (m) | Annul Rainfal (mm) |
|---|---|---|---|---|---|---|
| 0935 | Majhimtar | Dhading | 27.8055043 | 84.7054395 | 327 | 2711.643 |
| 0863 | Sakhar at Tanahun | Tanahun | 27.86741111 | 84.54938056 | 288 | 1568.736 |
| 1002 | Arughat (rainfall) | Gorkha | 28.043503 | 84.812925 | 518 | 1867.464 |
| 100401 | Nuwakot | Nuwakot | 27.914975 | 85.16463889 | 966 | 1901.46 |
| 0840 | Machhi Khola | Gorkha | 28.23155 | 84.874304 | 550 | 1977.612 |
| 1057 | Pansayakhola | Nuwakot | 28.02851111 | 85.10623056 | 1982 | 3066.324 |
| 0934 | Gajuri | Dhading | 27.803136 | 84.878054 | 430 | 1541.232 |
| 0928 | Shaktikhor | Chitwan | 27.72113611 | 84.58525556 | 333 | 2541.816 |
| 1038 | Dhunibesi | Dhading | 27.722878 | 85.164311 | 991 | 1443.036 |
| 0927 | Bharatpur | Chitwan | 27.6774101 | 84.4345037 | 216 | 2221.764 |

**Table 9: Soil metadata**

| DOMSOI | sand _ topsoil | silt _topsoil | clay _topsoil | OC _topsoil |
|---|---|---|---|---|
| Bd | 32.7 | 30.3 | 37.1 | 3.28 |
| I | 58.9 | 16.2 | 24.9 | 0.97 |